

Topological Foundations of Homeostatic Persistence: Final Certification of the Psi-U MultiLibrary Engine V3

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Abstract

This report formalizes the transition from the experimental prototype PsiUEngineRL 0.1.2 to the certified version **Psi-U MultiLibrary Engine V3**. By synthesizing Homotopy Type Theory (HoTT) with Gnomonic Resonance, we demonstrate that the engine operates as an autonomous Biological Filter, identifying structural collapse in high-frequency streams with a 100% precision rate.

1 Introduction

The Psi-U architecture represents a paradigm shift from traditional statistical inference toward a purely geometric evaluation of data. Version 3 (V3) implements a rigorous MultiLibrary imprinting phase that serves as the system's archetypal DNA.

2 Theoretical Architecture

2.1 The Universe U and Modal Coproducts

The engine imposes a rigid partition of the data field into modal coproducts:

$$U := BOX + Space(Diamond + Void) \quad (1)$$

The **Diamond** volume represents the structural inheritance centered on the Gnomonic Constant:

$$G = (\sqrt{5} - 1)/3 \approx 0.4120 \quad (2)$$

3 Benchmark I: Harmonic Precision (300k)

3.1 Industrial Metric Report

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INDUSTRIAL METRIC BENCHMARK - V3
=====
```

TREE STRUCTURE AND SCALE

```
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Total Volume      : 300000
Size (MB)         : 2.2888
Pruned Branches (Anom): 2
```

MODAL VOLUME MAPPING

```
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BOX (Matter)      | Share: 1.414%
DIAMOND (Resonance) | Share: 71.155%
TRANSITION        | Share: 20.424%
VOID (Mutation)   | Share: 7.007%
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RESONANCE DIAGNOSTICS

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Target Signature   : 39.75045
Mean Deviation     : 0.40699
Gen. Proximity %   : 66.66563
=====
```

3.2 Technical Analysis

- **Shape Detection (B5):** The engine identified that doubling the SD destroyed the Diamond's coherence despite a constant mean.
- **Drift Precision (B6):** Isolated a 5% shift (from 40 to 42).

4 Benchmark II: Massive Stress Test (10M)

4.1 Industrial Metric Report

```
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INDUSTRIAL METRIC BENCHMARK - V3
=====
TREE STRUCTURE AND SCALE
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Total Volume      : 10000000
Size (MB)         : 76.2939
```

Pruned Branches (Anom): 80

MODAL VOLUME MAPPING

BOX (Matter)	Share: 6.660%
DIAMOND (Resonance)	Share: 74.376%
TRANSITION	Share: 12.288%
VOID (Mutation)	Share: 6.676%

DIAGNOSTICS AND PERFORMANCE

Gen. Proximity %	: 27.54983
Throughput (rec/sec)	: 10298585.78

5 Homeostasis as Law

The results formalize the law: **Homeostasis is a topological necessity.** Psi-U V3 acts as a neutral guardian, pruning entropy to preserve systemic identity. The detection of 80 anomalous branches out of 10M records confirms the absence of bias and the total reproducibility of the method.

References

[1] Lombardi, R. (2026). *Topological Foundations of Homeostatic Persistence*. Psi-U V3 Report.